

- 3 (a) Explain why gravitational potential is a negative value for an isolated mass.

Gravitational potential at infinity is defined to be zero [B1]

Since, gravitational forces are attractive, [B1]

the force exerted by external agent to bring the test mass from infinity to that point (without a change in KE) is opposite in direction to the test mass's displacement. The work done by the external agent is negative and by definition of gravitational potential it will take on a negative value. [B1]

.....[3]

- (b) A satellite can orbit the Earth along an east-to-west direction (known as a retrograde orbit) as well as along the west-to-east direction (known as a prograde orbit).

- (i) A satellite is launched in the west-to-east direction from a launch pad on the Equator to the geostationary orbit.

Explain why this launch direction is preferred.

same direction as Earth's rotation about its own axis so satellite begins launch with some speed in correct direction [B1]

satellite already has some kinetic energy so less fuel needed to raise the gravitational potential energy [B1]

.....[2]

- (ii) The Earth may be considered to be a uniform sphere of radius 6400 km with its mass of 6.0×10^{24} kg concentrated at its centre.

Show that the geostationary satellite is 3.59×10^7 m above the Earth's surface.

gravitational force provides centripetal force [M1]

$$\frac{GMm}{r^2} = mr\omega^2$$

$$GM = r^3\omega^2$$

$$r = \sqrt[3]{\frac{GM}{\omega^2}} = \sqrt[3]{\frac{GM}{\left(\frac{2\pi}{T}\right)^2}}$$

$$= \sqrt[3]{\frac{(6.67 \times 10^{-11})(6.0 \times 10^{24})}{\left(\frac{2\pi}{24 \times 60 \times 60}\right)^2}}$$

$$\left[= 4.23 \times 10^7 \text{ m} \quad (4.2298 \times 10^7 \text{ m}) \right] \quad [\text{M1}]$$

$$h = r - R_E = r - 6400 \times 10^3$$

$$= 3.59 \times 10^7 \text{ m} \quad [\text{A0}]$$

[2]

- (iii) A satellite of mass 1000 kg is in geostationary orbit. Find its total energy.

$$\begin{aligned}
 E_{\text{total}} &= E_K + E_P \\
 &= \frac{1}{2}mv^2 + \left(-\frac{GMm}{r}\right) = \frac{1}{2}r\left(\frac{mv^2}{r}\right) + \left(-\frac{GMm}{r}\right) \\
 &= \frac{1}{2}r\left(\frac{GMm}{r^2}\right) + \left(-\frac{GMm}{r}\right) = \left(\frac{GMm}{2r}\right) + \left(-\frac{GMm}{r}\right) \quad [\text{M1}] \\
 &= -\frac{GMm}{2r} \\
 &= -\frac{(6.67 \times 10^{-11})(6.0 \times 10^{24})(1000)}{2(4.2298 \times 10^7)} \\
 &= -4.73 \times 10^9 \text{ J} \quad [\text{A1}]
 \end{aligned}$$

total energy = J [2]

- (iv) Atmospheric drag is very low but nonetheless present at the height where geostationary satellites orbit.

Explain, in terms of energy, the impact of atmospheric drag on the subsequent trajectory of geostationary satellites.

Work done against drag so total energy decreases [B1]

GPE decrease so satellite lowers in height
KE increase so linear speed increase } [B1]

satellite spirals towards Earth with increasing speed [B1]

.....

 [3]

[Total: 12]

